

ActInf Textbook Group ~ Cohort 7 ~ Session 23

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Hey, welcome fellows. Um, we're in the second discussion on chapter 9 for cohort 7. So, we could go to any part of chapter 9 or jump in wherever you prefer. What about the provocative first sentence? Ultimately, the models described in this book are only useful if they can answer scientific questions. Uh yeah, I liked uh this is uh old chapter because of its uh uh well because of the way that it um inverts everything for the sake of uh uh applying this to science. I think that that's it's interesting the cognizance of it all to realize like uh maybe what's been done so far is helpful as a model but that doesn't really um uh that's not the same as let's say a scientific um I guess result and so but to turn it around and say it is quite impressive if you can uh do it backwards so that if you can if I understood this correctly If you can suppose that there's this model in a subject or maybe a group of subjects and then you can um given that supposition then maybe figure out what the parameters should be um or what the what the results should be and then work back to say okay well if the if these are the results then what are the initial beliefs you know between different um participants in a population and then if you can do that then it's interesting oh that you have a a you know potentially a scientific explanation of how these different uh individuals in a population differ and why and to be able to show the helpfulness of the model and as explanatory and then you can correlate that with other information. So that's how what I took from this. Yeah, great points. It has a lot to do with what makes a scientific account like any let alone a satisfying or useful one. And what is the relationship between empirical observables like the brute fact of the data points and inferred latent states like causal attributions. Mhm. or unobserved features which are derived from empirical data but derived from empirical data does a lot of work because you could derive in in so many different directions from empirical data. You could say that this art piece was derived from the weather data and there's so many degrees of freedom with what weather data you took and and what art piece you resulted in that somebody else might not be able to look at it and know that you use weather data at all. Right now, I'm interpreting this statement

to mean uh I'm interpreting answer scientific questions to mean make testable predictions. That may not be everyone's interpretation, but I always thought models are only useful if they can make testable predictions. Yeah, that's a great point that's related to um like falsificationism and the idea that we create secondary hypotheses that could then be either assessed to be true or false under the logic like if what you learned from the answer didn't give any derivative product that could be deemed to be more or less valid. It was kind of a dead end answer because it doesn't it doesn't give any uh consequences. And then testable predictions is like the empiricist way of framing it and that can even be framed and staying within the actontology like answering scientific questions means reducing uncertainty about the scientific question. Oh, I like that. Now, alternatively, um, pointing at your the first part of what you just said, it's still useful if we're moved forward in our investigations, even if it doesn't have explicitly testable predictions. If we're still developing the field of inquiry, that's still valuable. And and uh when you say testable predictions, I think that that comes at it from both ends that are necessary in science because you want to have things related to data that's observed, you know. So if it's if it if there's no data involved and there's no observations involved, then that's a funky science. But then also you want to have a theory. Uh so you want to have a a model that is um meaningfully breaking down or decomposing or structuring um these variables or questions or ideas and so that you can look at relationships and so if you have the testable prediction it means that you have it from both ends and then you can apply it to a wider scope you know you can look at new examples etc but I think in the case of this chapter it's about predictions of predictions which is very um sophisticated because you're basically predicting um well that you can figure out what's in another organism's mind, you know, what they're predicting, let's say, and then and if you're if your predictions correct, you'll be able to do that for a whole variety of individuals to say, well, they're predicting these different things. And you'll be able to confirm that uh and then so the confirmation can also happen on both levels that you can do that for more and more individuals but given an individual you can also get better and better validation for like well what exactly is that uh belief or prediction that they're making. Um so it's sophisticated again pointing to the process nature the dynamic nature the approach that and and talking about it this way with chapter 9 being so grounded with empirical data it's kind of like the the mirror symmetry of the book. Chapter 1 and 10 are like these overview chapters. Chapter two is the low road. So when we when we when we got on that on-ramp, it was about building the model and the joint distribution and basian statistics. And now we're kind of exiting on an empirical low road because it's again reconnecting us

back to the observations. Like this is a setting that's very familiar to a behavioral scientist. And even when a paper doesn't uh like a behavioral science paper doesn't explicitly use these techniques doesn't use a palm dp doesn't calculate variational free energy still basically the setting of this metabasian um setup is there like here it's the temperature and the humidity in the laboratory and they're the uh experimentter is playing sequences of videos or they're presenting symbols on cards which are observations for the subject and then they're receiving the action output of the subject. And then this just does so much interesting work with the subjective because it is both subjective to them and it's the subject of the behavioral experiment. And it's like this is what a laboratory is like. And then this solid box is like the kind of the veil of unknowing for the behavioral scientist in terms of they can set up what's in the room. They can pass the stimuli to the subject and they can observe what comes out with biometrics or or with whatever kind of measure and then what's inside is proposed. So this structure it is is a really parsimonious way to just look at the behavioral scientific setting. And it turns out it has a lot of structural congruence with the metacognitive phenomenology like an observer observing with within an [Music] organism. But that goes into kind of a different direction than this like more behavioral laboratory very practical and and this and this does get really practical and the by number or by citation count most of Fristan's papers may be this kind of like this many subjects were put into the fMRI for this many minutes exposed to these stimuli here was the statistical power here were the the um here was how the experiment happens. Um, so this gets to the most practical statistical power calculation like parts of doing behavioral science which kind of in in the um skarian area like BF Skinner that without going too much in this that's kind of the whole debate. Does that deny mind? Does behaviorism exclude mind? Um, and the the the cover of the book told us that we were going to be in for mind, brain, and behavior. And we saw brain in chapter 5 with the different neural systems. Chapter 9 really goes after behavior. because it's not making psychological assertions um or or um philosophical assertions. So mind is obviously a a really complex and rich area with the first person and the second and the third person all these and all these different traditions and and our own uh firsthand experiences of mind. Uh but nine zeros in on behavior which in a way gives the least contentious of the three. That statement might be contentious to some but the neural accounts somebody might think well with um molecular reductionism and genetics and systems biology and deep learning we're doing better than chapter 5 on brain accounts. But in terms of the structure of the uh environment, this is really like a gem tucked in at the end of the book. I like that perspective. Uh there are many traditions that say reality is

what happens. That's behavior. Well, and I think for this chapter, I don't know if I'd entirely agree, but maybe I understand u but with Daniel, but uh uh because a lot of these uh examples at the end have to do with psychological um analyses like let's say hallucinations, right? Or delusions, right? So that's partly behavioral, but that's partly their internal models, right? like and and I think one of the challenges with these experiments just thinking about u what it would be like to do one is that with active inference in its full glory it has u two outcomes you could be updating your environment or you could be updating your belief so then if you're doing an experiment you want to capture that I think that'd be like the whole glory of it you know you have to be on top of two different uh channels you know the external like the the behavior you to be able record, oh, you know, behavior updated the environment, let's say, but you also want to be able to capture, well, oh, here they updated their belief. So, that already is kind of like a double thing. I don't know if that's I'm great great points and and and challenging ones. So yeah from the first half of the book we have this two ways to to dissipate free energy with update behavior or update belief. Now from the behavioral uh researchers perspective everything coming out of that box is observed behavior whether it's an EEG trace um or whether it's a self-report or whether it's a motor behavior. So from the outside everything that we're observing is behavioral and basically everything that is internal is being modeled as a belief. That doesn't mean that it's like a um propositionally held sentiment. Mhm. But it's being modeled as a basian belief which undergoes updating. So you're totally right that if we want the full active inference account, we do want to be able to describe both the updating of behavior, but this is the updating of observables. So if we only wanted to describe behavioral updating, we could just do descriptive statistics on sequences of actions. And there might be a situation where that's totally fine. But as soon as the behavioral researcher starts to suppose some latent factor that wasn't behaviorally observed but is um abduced to be relevant for the generation of sequences. They do start to kind of swim upstream into cognitive states which do get into mind like that's where this where those um examples in the table basically go from they're grounded in self-report or secade um or decision or uh game theory games or trust games and then um latent states are proposed which make their way kind of up the low road towards the psychological and the cognitive. Mhm. Um but maybe just to emphasize like what I think is scientific about this chapter u compared to what we've seen before like if if somebody comes up with a computer computer model using active inference principles to explain like how a hand reaches out and manages to gracefully pick up a chest piece and move it, right? like so that there's this uh model of how the muscles gracefully uh relax or or tense

up and they do all that. That model is uh interesting. It could be very slick. Uh it could be it would be it's always impressive but it's not convincing that that's scientifically you know really relevant in a sense because there could be many kind of slick ways to model that. There could be many biological processes that explain that but like what this chapter is uh portraying I think or suggesting it's it's saying look if you have let's say a group of individuals and you're able to model their behavior but then you're able to show look some of these individuals are applying a different strategy they're having different beliefs they're getting different results you know because see so that's a very discreet great difference. You're able to say like in these cases it's one way, in these cases it's another way. So scientifically like that becomes um that for somebody like me that's very convincing that oh that may well be a you know the right explanation. Maybe I'm gullible to things like that but but um I find that more uh yeah like let's just say we do the rat and the teammates like the chapter seven experiments. Mhm. So if we just wanted to say we run 100 mice for 100 replicates each. So if we just wanted to say okay we're making a leaderboard which one got the highest percentages. Mhm. We we just have that raw data tabulatable from description or if we wanted to say um just a simple correlation between the temperature in the room and the success rate which mice did better when it was hot. These are still basically you take two observables, you know, we have the observable temperature, observable successes, and we just basically there's our answer. So that sort of doesn't appeal to any latent variables. But then like you're saying, if we were going to want to go beyond, so then we say, well, how how likely do we feel that the leaderboard would be the same on the next Mhm. replicates or if the condition conditions changed or we want to study transfer learning into a different setting. That's where I mean why go through all this work of making a specifying a generative model with cognitive states doing model inversion to get error bars on unobservables when if we just have the observables let's just tabulate them. Mhm. And so again, that goes to sort of the depth and type of scientific account that one seeks. If we just want to summarize observables, this whole apparatus could be overkill. Like if we were just trying to say we just want to know which cars are breaking the speed limit. It's just like as soon, okay, it's going over the speed limit. Boom. There's the rule-based observable. Mhm. But if you wanted to say we want to be inferring something about the attention level or the cognitive state of drivers. Mhm. And maybe want to have one model that's just speed. One is just speed and exposition or something like and they have all whole portfolios of models, portfolios of cognitive states, noisy or patchy data. So it's a much more flexible modeling situation. Mhm. to be able to go from a custom sensor array to a

custom cognitive unobservable suites and then treat those unobserved inferred variables as downstream as observations for a downstream process. like this could be a risk factor, but the but a risk factor is never going to be a measurable feature of the world. there could be an observation that's a good proxy for risk or something like that, but any kind of summary statistic or um risk estimator, there's just all kinds of stuff that seems pretty obvious to want, like which one of these buildings is going to be most likely to have this happen, but that will never be revealed from just munching the data from observables down. But even even like with a risk factor like risk factor is generally not a satisfactory causal explanation. uh well I mean it's maybe helpful and practical in some ways but but if you really want to know the cause right like um there's usually something direct and I think what this is offering uh is to say that like if different individuals have or organisms have different um generative models they're going to have different causes to their behavior yeah again to like what makes a scientific account what if we Okay. Well, this person uh this building was a high risk of falling. Mhm. Why did it fall? It had a high risk of falling. It's like, well, that's not really an explanation. That might be very useful that you were able to have a a a calibrated risk assessments or which one of these cars is likeliest to speed. So that gets into the testable useful hypothesis, but it's kind of like that's an that's a account in the investigator's mind, a a utility oriented account, but that's not even being proposed to be an actual natural account of like why the building fell, right? Which might relate to like the total gravitational load and like the supportive ability or something like that. But the design, the design and the materials, right? Also, Andreas, I think you point to why uh active inference crops up in so much of the embodied and active mind literature. Uh whether the internal models are updated or in action is the is what's being updated. They can't be separated. They they interact with each other. Yeah. Like the the moot or the trivial system is just behavioral. So then there's no need to abduce a latent state. So it's kind of like it can't update beliefs. It can only update its action. Yeah. But once you start getting into anything um another important piece is is the uh introduction of the basian approach. So if we go out we we do the mice studies and like 75% of them succeed and you say well what's your best estimate on mice succeeding not how many mice succeeded in your experiment but what is your best estimate on mice succeeding in this if you were to run it next week or if you were to have another student run it or something like that. The the frequentist answer is 75. You know we flipped the coin 75 100 times we got 75 heads. So our best estimate is 75 and that's like the maximum likelihood estimator. Whereas with basian statistics we might have like a metaanalysis of similar studies. Mhm. And say like

we have a pretty strong belief that it's about 50%. So then on a given day when we get a run and it gets 75 with attention full attention to the experimental results we could say we now believe that it is 75. we believe the situation is rapidly changing and that now mice solve it 75% of the time or and say this is just this many sigma it's a three sigma events one in 700 times we expect to see that we have now seen it one in 700 times and we're not updating from 0.5 at all or it could be anywhere in between with like yep now our best posterior estimate is 52%. So that's the part of this whole work of specifying the base graph and doing this this bidirectionality of like generating synthetic data or receiving empirical data is so that we can explicitly include a prior belief term because the just the much the description you know much the data style describe the observations you don't really get a chance to include your prior Whereas Bayesian statistics is basically just including your prior as an observable. Mhm. For the analyst and then then making choices about how much to weight the prior versus the data coming in. Mhm. Which again is that kind of reflexive element because it's like oh yeah how much how precision waiting on how much to do prior updating given the observation is the Bayesian statistics question. Mhm. And descriptive statistics lends itself, you know, not not to weld things together too tightly, but that lends itself to the p value accept or reject falsification. Mhm. Whereas the Bayesian epistemology and and there's many like writings and explorations on this lends itself to more this continuous probability space updating style. It's like well is the sun going to rise tomorrow or not? you know, whatever. Will we come to see the sun tomorrow? Mhm. And the frequentist could just say yes. Or, you know, I failed to reject that it didn't. It's like all those kinds of expressions as opposed to just having a very high posterior belief, but a Bayesian and and you could make a Bayesian model that does give a discrete answer, but if you don't constrain it that way, it's going to stay in the continuous probability spaces which have more agility. Mhm. because it's it's like okay well we shut we shut the door on that hypothesis. It was it was um it was falsified. But then part of the um sometimes famous nature of the frequentist test is you're presented with two hypotheses and then either there's a rejection or a failure to reject one and then the failure to reject is not the same as the acceptance. And the rejection of one isn't the same as the acceptance of the other because it can all be down this super contrived sequence of assumptions. So it could be like we tested to see whether eating eggs made a difference in cholesterol at this p value. We didn't reject the null that it didn't. But that's like I don't know if it's like the contra-positive or something like that. But it's not the same as saying that it doesn't. It's just saying that that study wasn't powered or or sensitized to detect the rejection, which is which has a huge overlapping territory with it

actually not making a difference. Mhm. So all all of this is kind of like in practice, they might all give the same results. If you're just testing for the difference in temperature between two rooms and the difference is big enough and clean enough, probably a huge number of methods are going to get you the right um actionable information. But this is getting towards a more sophisticated accounting that like helps us embody all of these nuances that that that do come into play. This brings us back to the great power meaning uh necessitating careful model construction. uh sort of like in our other session, it'd be worthwhile. I'm looking for a course on modeling. Uh the philosophy of models. Well, I would talk about the model I'm thinking of that's uh uh this chapter kind of provoked me to think about uh because I'm I like to model human experience. Um I've been working on that for decades and and so our active inference is um attractive um and um but and so active inference is quite universal but but maybe not completely. So an example of the type of problem I thought I would do I would like to know what are the languages by which uh for a human being uh things come to matter or u let's say a language for how does meaning arise or how does a person believe like an event happened you know I think that there's maybe some basic cognitive languages for that so I needed data for that so the data I thought I would study uh is u triangle geometry uh that you can have triangles of different shapes and then you have these concepts of triangle centers which could be the simp a simplest one is what's called an in center. You draw a circle inside a triangle and you make the circle as big as possible. The center of that circle will be called the in center of the triangle. But you could do it with a center on the outside. And there's all these ways that you could um make points given a triangle like you know you look at the medians you you intersect them things. So but there's not that many ways. And I suspect if you if I study that and there's an encyclopedia of triangle centers and if I study how it builds it builds a language a language like this or maybe any language will probably have six underlying actions uh which relates to let's say six cases uh Daniel gave a um a talk about u cases in language let's say so something like that these type of prepositional ways of thinking um that nouns are let's say used in a language like Lithuania or Russian or so on or Latin. So um the point being that okay as far as active inference goes it's kind of interesting that it's possible to think of there is this kind of like coordination between two types of action. One is let's say a physical continuous action like making a circle bigger. The other one is saying there's like a conceptual point to say I want to have a center of a circle. So you do actions like you make a circle bigger. It's not positioned well. You kind of bump it over. You keep doing you get you do a chain of actions. You end up with something like that's a center. Let's say that could make sense for a

generative model saying this circle needs a center that I could focus on. And maybe you can redo it for different triangle shapes and do different types of triangle. You get this whole kind of activity for language. The problem with that based on this chapter is that yeah, but is that ever going to be science? you see like is that is that ever going to solve a scientific issue because you're just um studying um you know these triangles it's not in a certain sense it's not really data from the real world in terms of experimental data on the other hand it kind of is because you have this whole you know encyclopedia of triangle centers people came up with what does active inference say about an encyclopedia like that like how to make sense of it like so if there was a modeling language or approach that could go into that data and say hey this is how to un this is what to do to understand it as a language now so the one thing I wanted to just conclude with is that um what this chapter got me thinking though is that in order to make it to the next level where it's kind of more observable or it's more meaningful you kind of need the system or the game to um maybe like be communicating with another individual. So like let's say there's two people playing this game of u the triangle or maybe like you know you're just communicating with yourself you're replaying you know you're trying to get good at this you're trying to make sense of you know what is a triangle uh what helps me understand triangles see so when you model that and how those games are related and I think the reason being that like in the science of this what's impressive is when you have two different people with two different generative models and they get different behavior so if you do that in a single person like with this type of game saying well you can go into different modes or have different beliefs or they can change or you can do that in communicating with somebody else then I think that becomes the kind of thing we're trying to be able to model yeah very interesting yeah Stephen please I think that's the essence of the human mind that there are two generative models that are compared with each other there's an online sensory input and then there's the internal representation of that sensory input and those get compared with each other and it's that comparison I think in in the higher level cortical areas that is this really complex motor behavior that we call thought sense of self and motivation. I I I wonder so there's that. Uh I I wanted to ask if you were using meaning in the sense of concept like the formation of meaning is the formation of concepts. I think there's a whole literature in what concepts. Well, of course, there's literature in what meaning is also, but I'm always approaching this from the standpoint of the actual architecture of the brain and the body and how this is implemented uh you know neurons or basian units themselves and then they form basian circuits and and it's the dynamic processes you know with these frequenc of uh uh these

activations with different frequencies that allow signals to get through or not, you know, that amplify or cancel signals. And you know, it's it's so hard for me to constantly translate basian terminology like beliefs into a structural terminology like neural circuit. So um what I find very helpful what you said was like that you see generative models at both levels and I really hadn't thought about that but maybe that's the way to go. Um I have this model of the three minds I call it. Uh so there there's a mind that knows the answers uh which would be what you call like the sensory um and it knows them unconsciously but then you have a mind that uh uh does not know but it consciously does not know. So it's just like full of questions. It's full of variables like every word is a question like so like the word horse really means like what is a horse and then the other mind gives you the answer what the horse is maybe gives you an image or something. So so in terms of meaning arising it's like basically you have these two minds on autopilot you know like one is just um one is just feeding you all kinds of imagery like you're watching television the other one's just kind of chattering away you know and they just do their own thing. Um, but you have a third mind that's investigatory that tries to connect them two, you know, with each other to kind of like um balance them so that the mind that knows and the mind that does not know, they're able to say the same things in their different ways. Cuz like the mind that knows, it's like a 100 billion neurons are telling you one thing, but there's like a 100,000 concepts that that same information is being reduced. And now so it's like this distinction between the brain and the mind, you know, like that is that neural well that's maybe not a very helpful way to look at it. It's just some kind of conceptual structure out there. Um but there they need to be linked. So there's a third mind that kind of helps to point punctuate or um you know adjust ratios, parameters, whatever to kind of uh emphasize like to weave them together. So on the one point of view like with the conceptual mind there's a certain point where like it's babbling and all of a sudden yeah but this is meaningful and that is meaningful for this reason and for that reason like for example this point at this circle is going to be meaningful that we're building. Uh whereas like the the points on the circle aren't meaningful at all but in terms of mattering like it's it matters to get the circle to the maximal size that matters. That's a task to perform. So, I like that little language because you can see the difference between the two. Yeah. I think the third mind falls out of the other two. It's a consequence of the interaction between I really like your framing there. Yeah. And so, it's both a consequence, but then maybe it's steering it. You see? So, like we have a will. It's deliberate. It's cognizant. So, it's a it's ambiguous it's ambiguous, right? Like which is it in the process? It changes. One could say it changes the synaptic weights but the uh uh there there's nothing

static. Everything is constantly changing and when they meet the interaction changes itself in the process of interacting. Think of I always think of it in terms of a synapse changes itself in the very action of transferring neurotransmitter. the vesicles have proteins on them that merge with the synaptic membrane and now it's different than it was before. Yeah. So is that is that a helpful way to look at neuro is is active inference and generative model I mean generative models are they a helpful way of looking at the neuronal neural systems themselves. It's a it's rich enough to apply to the neural systems themselves. Absolutely. Yeah. It's very it's quite appropriate I think. uh there I think there that's where it's really helpful to look at the SPM work which is a whole toolkit ranging from like spike and fire models of single neurons to neural mass probability models to dynamic causal modeling all within this like ultra chapter 9 empiricism with the sex fusion of the EEG fMRI data Um so this and then in a way the the scientists model can be seen as like a symbolic second mind of the primary embodiment of the mouse. But let's say it was like an alien creature and it was it was or it's an unknown creature but it h it has an embodiment. It has a firstness. Mhm. And then this is a symbolic projection onto that firstness from a pretty small diagrammatically defined graphical toolkit. Mhm. And then there's a to your point about the communication then there's a two agents. It's kind of like that. Then the minds is kind of like the the experimental setup and the experimentter. Mhm. And then there's the investigation is in the interplay. And then you talked about communication of generative models across agents. That's like the synchronic like joint. That's that's like the space between two at or or more at once. And then there's um replay and and pre-play which is like diarrhonic but that so that's like one self but but it's like is it a different self right? If it's a but you know just legally speaking it's the same person. Mh. But then that's like also kind of like a space between some extent actuality which is like the 100 million neurons. Mhm. And then some symbolic condensation and then some something happens in the space between which in the basian updating moment is given a causal power. Or you could just take the the basian posterior and just it could be treated as epiphenomenal like you could just continue to to collide streams of priors and observations. Mhm. And then just slough off the posteriors. What do you mean by a phenomena? What does that mean? Like you'd calculate a posterior and discard it. Like the the posterior is never given a causal re-entry into the system. Oh, okay. Like what what people arguing that that consciousness is epiphenomenal, which is to say like it arises but there's no causal arrow back like it's like an it's like a side effect but not really relevant. It's not a participant. Yeah. And then the question is like you know without you know dart and all that but how do you have something that re-

enters causally is the third mind just like the sparks flying from the interplay but then but do those sparks you know can they or you know you know if weather whether it's that all those things that that you know do they re-enter to modify either neural activity or symbolic activity and and so I'm writing this paper to about this model and and u some of the things I write about uh and I quote the textbook uh but for example there's an interesting paper about a window that says you know you have pre you have a stimulation of the brain let's say you have preconscious uh uh time but at let's say from 250 milliseconds to 700 milliseconds there's this rather long window of conscious axis is what they call and what then what they noted with the paper they did different situations where you could do task related or uh taskfree uh you know you're listening to music let's say maybe you're supposed to be hearing look hearing for something or maybe you're not but but they they look at the conscious axis and what they notice is that um the window stays open for longer than you would think like there's room for playing around of the signal there's room for it to settle down so with regard to the three minds then that's very compatible in the sense that your first mind these this preconscious stage can suggest candidate action uh sensations it can give you it can give you information like that then your conscious mind is globally trying to figure out you know how to integrate this what to do but the third mind it's like acting as a break it's saying wait you know don't rush don't rush I'll tell you when to hardwire it when it's all right so the point being that the prolongation suggests that there could be something that's prolonging this, you know, like that's that's that's ex, you know, fighting against early u early resolve resolution. But furthermore, you can have an idea go into your brain before you're conscious of it of what you're going to do. It's just a candidate, you know, decision. So you get this paradoxical thing that oh what you thought you were going to do you already knew quote unquote before you but actually it was just a candidate you know that got confirmed. I think that's the way to read it. I like that which then is affected by experience. the more that happens, the more likely it's Yeah. But all three zones like the meaning changes as you go through those three zones of you know it was a candidate it be got kind of like worked out and then it was affirmed let's say and and I think that points to a threshold effect uh where if there is this practiced development of openness to intention then if consciousness can be ever be considered as anything other than epiphenomena it is in the realm of intention. uh creating as you say uh a bias toward a particular candidate class of yeah you can have a bias right and so that's a way of self affecting yourself is to have biases and so so I quote chapter five uh about the basil ganglia because I say there's a lot of evidence I think Mielchris Ian Mielchris wrote a book about left hemisphere versus right hemisphere

so you get a lot of this you know a champion and for both but I'm saying it's deeper than that. If you go to chapter 5 about the basal ganglia uh even in arthropods uh you have this uh uh distinction between u releasing and allowing the action as opposed to not allowing it. There's this you know you have to play with both and I think that that's a sign of consciousness that uh you know you're able to do that. Uh and so dopamine is what triggers that and dopamine goes way back. So that's just the mechanism. Another one I cite is uh uh immune systems. You have two versions. You have um the um the one that's like inborn, innate I guess they call it, then the one that is able to learn. And so the question is how do they not conflict? So I think there's got to be some kind of harmony between them, harmonizer between those two immune systems. You talk about another principle of biological function. uh the the best control system is one that is both excitatory and inhibitory in constant right you want both right and so I I have this quote I I I said this thing the second mind it avoids the territory because there's things you can't learn from you know you'll just die so you want to avoid the territory and embrace the map uh and so that's why you want to for things you know you you know like the unlucky mind there's a mind when things are unlucky you don't want to try your luck we have a lucky mind and an unlucky mind so okay thank you we'll end it there see you